

Curriculum vitae

Federico Binda

WORK EXPERIENCE

Research Professor

Simons Laufer Mathematical Sciences Institute (SLMath), Berkeley (CA)

August 2026 – December 2026

MOTIVIC HOMOTOPY THEORY: CONNECTIONS AND APPLICATIONS

Associate Professor

Università degli Studi di Milano

October 2024 – present

MATH 02A, ALGEBRA E GEOMETRIA

Assistant Professor tenure track (Ricercatore Tempo Determinato B)

Università degli Studi di Milano

September 2021 – September 2024

MAT 02/ALGEBRA

Junior Assistant Professor (Ricercatore Tempo Determinato A)

Università degli Studi di Milano

June 2019 – September 2021

MAT 02/ALGEBRA

Postdoctoral researcher

Universität Regensburg, Germany

Sep 2016 – May 2019

Arbeitsgruppe Prof. Dr. Moritz Kerz

SFB 1085 HIGHER INVARIANTS

Wissenschaftlicher Mitarbeiter

Universität Duisburg-Essen, Germany

Sep 2015 – Aug 2016

Arbeitsgruppe Prof. Dr. Marc Levine

DFG SPP 1786 HOMOTOPY THEORY AND ALGEBRAIC GEOMETRY

EDUCATION

Doktor der Naturwissenschaften – Mathematik (Ph.D.)

INSTITUTION Universität Duisburg-Essen, Essen (Germany)

10 August 2016

DISSERTATION TITLE Motives and algebraic cycles with moduli conditions

GRADE *summa cum laude*

SUPERVISOR Prof. Dr. Marc N. Levine

Master 2 Recherche, Mathématiques Fondamentales et Appliquées

INSTITUTION Université de Paris Sud, ENS Paris, École Polytechnique

22 November 2012

NOTE Double degree program (ALGANT project, Erasmus Mundus Master)

Laurea magistrale in Matematica

INSTITUTION Università degli Studi di Milano

18 July 2012

DISSERTATION TITLE Hodge-Tate decomposition theorem for Abelian Varieties over p-adic fields

GRADE *110/110 e lode*

SUPERVISOR Prof. Ahmed Abbes, CNRS - IHES (Bures-sur-Yvette, France)

Laurea triennale in Matematica

INSTITUTION Università degli Studi di Milano

22 July 2010

DISSERTATION TITLE Units in quadratic number fields

GRADE *110/110 e lode*

SUPERVISOR Prof. Massimo Bertolini

LONG RESEARCH VISITS AND VISITING POSITIONS

- **The University of Tokyo, Tokyo, Japan**
Graduate School of Mathematical Sciences, Visiting Professor

Spring 2027 (Planned)
- **Simons Laufer Mathematical Sciences Institute (SLMath)**
Research Professor. Program “Motivic Homotopy Theory: Connections and Applications”

Fall 2026
- **Isaac Newton Institute for Mathematical Sciences, Cambridge, UK**
Member. Scientific program “Algebraic K-theory, motivic cohomology and motivic homotopy theory - II”
<https://www.newton.ac.uk/event/kah>

3 May – 29 July 2022
- **Center for Advanced Studies (CAS),
Norwegian Academy of Sciences and Letters, Oslo, Norway**
Research Fellow. Scientific program “Motivic Geometry”
The program was held entirely online due to the COVID-19 pandemic
<https://cas.oslo.no/research-groups/motivic-geometry-article3332-827.html>

Sep 2020 – Jun 2021
- **Isaac Newton Institute for Mathematical Sciences, Cambridge, UK**
Scientific program “Algebraic K-theory, motivic cohomology and motivic homotopy theory”
The program was postponed due to the COVID-19 pandemic
<https://www.newton.ac.uk/event/kah>

15 Mar – 22 May 2020
- **Tata Institute of Fundamental Research (TIFR), Mumbai, India**
Research visit
Work in collaboration with Prof. Amalendu Krishna (TIFR)

13-31 Jan 2019
- **Tata Institute of Fundamental Research (TIFR), Mumbai, India**
Research visit
Work in collaboration with Prof. Amalendu Krishna (TIFR)
Financed by TIFR and Universität Regensburg (DFG-SFB 1085 - grant Prof. Moritz Kerz)

15 Jan – 11 Feb 2018
- **Hausdorff Research Institute for Mathematics (HIM), Bonn, Germany**
Hausdorff Trimester program “K-Theory and related fields”
Organizers: Guillermo Cortiñas, Hélène Esnault, Christian Haesemeyer, Holger Reich, Jonathan Rosenberg

08 May – 31 Aug 2017
- **Institut Mittag-Leffler, Djursholm, Sweden**
Spring term research program “Algebro-geometric and homotopical methods”
Special thematic trimester
Financed by Institut Mittag-Leffler
Organizers: Eric M. Friedlander, Lars Hesselholt, Paul Arne Østvær.

26 Mar – 30 Apr 2017
- **Tata Institute of Fundamental Research (TIFR), Mumbai, India**
Research visit
Work in collaboration with Prof. Amalendu Krishna (TIFR)
Financed by TIFR and Universität Regensburg (DFG-SFB 1085 - grant Prof. Moritz Kerz)

1 Nov – 10 Dec 2016
- **Tata Institute of Fundamental Research (TIFR), Mumbai, India**
Research visit
Work in collaboration with Prof. Amalendu Krishna (TIFR)
Financed by TIFR e Alexander Von Humboldt foundation (grant Prof. Marc N. Levine)

15 Sep – 10 Oct 2015
- **Tata Institute of Fundamental Research (TIFR), Mumbai, India**
Research visit
Work in collaboration with Prof. Amalendu Krishna (TIFR)
Financed by TIFR e Alexander Von Humboldt foundation (grant Prof. Marc N. Levine)

1-19 Sep 2014

TALKS AT INTERNATIONAL CONFERENCES

- | | |
|---------------------|---|
| <i>June 2026</i> | <p>Fundamental Geometries
“On the p-adic deformation problem for the K-theory of semistable schemes”
University of Bologna, Italy</p> |
| <i>October 2025</i> | <p>Rencontre ANR DAG Arts
“Derived logarithmic geometry” (3 hours mini-course)
University of Angers, France</p> |
| <i>October 2025</i> | <p>Higher Invariants: interactions between arithmetic geometry and global analysis</p> |

	“Relative Hyodo-Kato cohomology” SFB Closing Conference. University of Regensburg, Germany
August 2025	Workshop “Algebraic K-Theory” “ p -adic deformation of algebraic cycles and the log Bloch-Esnault-Kerz fiber square” MFO Oberwolfach, Germany
April 2025	Conference “Recent Developments in Algebraic K-Theory” “Infinite root stacks and the Beilinson fiber square” University of Warwick, UK
March 2025	Conference “Motivic homotopy theory 2025” “Infinite root stacks and the Beilinson fiber square” University of Regensburg, Germany
July 2024	Conference “Algebraic K-Theory and Arithmetic” “Log prismatic cohomology, motivic sheaves and slice filtrations” Bedlewo, Poland
March 2024	Conference “Motives in Mainz” “Prismatic cohomology, motivic sheaves and comparison theorems” University of Mainz, Germany
February 2024	Workshop “Incontri di geometria algebrica e aritmetica” “Prismatic cohomology, motivic sheaves and comparison theorems” University of Pisa, Italy
August 2023	Conference “1786 Abschlusstagung’ “Motivic monodromy and p -adic cohomology theories” University of Duisburg-Essen, Essen, Germany
February 2023	Conference “Motives in Tokyo’ “Motivic monodromy and p -adic cohomology theories” University of Tokyo, Japan
October 2022	Conference “Motives, quadratic forms and arithmetic’ “On the logarithmic Hochschild-Kostant-Rosenberg theorem” University of Lens, France
August 2022	Conference “Motivic Geometry’ “On the logarithmic Hochschild-Kostant-Rosenberg theorem” University of Oslo, Norway
July 2022	Conference “Harnessing motivic invariants’ “On the p -adic weight- monodromy conjecture for complete intersections in toric varieties” Satellite ICM 2022 ; University of Duisburg-Essen, Essen, Germany
June 2022	Workshop “Algebraic K-theory, motivic cohomology and motivic homotopy theory’ “On the logarithmic Hochschild-Kostant-Rosenberg theorem” Isaac Newton Institute, University of Cambridge, UK
May 2022	Workshop “Algebraic K-Theory” “On the p -adic weight-monodromy conjecture for complete intersections in toric varieties” MFO Oberwolfach, Germany
Sep 2020	Conference: Motivic geometry “Triangulated motives for logarithmic schemes” Center for Advanced Studies, Oslo, Norway (virtual)
Jul 2020	IHES 2020 Summer School Motivic, equivariant and non-commutative homotopy theory “Motives for logarithmic schemes” IHES, Paris, France (virtual)
Sep 2019	Regulators in Niseko “Triangulated categories of log motives over a field” Hokkaido University Sapporo, Japan
Feb 2019	International workshop on Motives in Tokyo “Semipurity for cycles with modulus” University of Tokyo, Tokyo, Japan

- Sep 2018 Conference **Motives in St. Petersburg: Motives and their applications**
“Rigidity for cycles of relative dimension zero”
Euler International Mathematical Institute, St. Petersburg, Russia
- Jul 2018 Workshop **Motives and algebraic cycles**
“Specialisation theorems for cycles of relative dimension zero”
Freiburg Institute for Advanced Studies, (FRIAS), Germany
- Mar 2018 Conference **Motives in Tokyo, on the occasion of Shuji Saito’s 60th Birthday**
“Rigidity for relative 0-cycles”
University of Tokyo, Tokyo, Japan
- Feb 2018 Conference **K -theory, $\mathbb{A}^1$ -homotopy and quadratic forms**
“Rigidity for relative 0-cycles”
University of Warwick, Coventry, UK
- Jun 2017 Workshop: **K -theory and related fields**
“Towards a motivic (homotopy) theory without A_1 -invariance”
Hausdorff Research Institute for Mathematics, Bonn, Germany
- Feb 2017 **International workshop on Motives in Tokyo**
“Zero cycles with modulus and zero cycles on singular varieties”
University of Tokyo, Tokyo, Japan
- Nov 2016 Workshop **Around motives**
Cycles with modulus and non $\mathbb{A}^1$ -invariant motives
Università degli studi di Milano
- Jun 2016 Workshop **Homotopical approaches to categories and geometry**
“Beyond homotopy invariance: on motives with moduli conditions”
University of Freiburg, Freiburg, Germany
- Apr 2016 **Generalizations of $\mathbb{A}^1$ -homotopy invariance in algebraic geometry and homotopy theory**
“Zero cycles with modulus and zero cycles on singular varieties”
Freie Universität Berlin, Zinnowitz-Usedom, Germany
- Oct 2014 Conference for Young Researchers in Arithmetic and Algebraic Geometry
“Algebraic Cycles with modulus and Applications”
University of Bonn, Bonn, Germany
- Sep 2011 Summer school **Cohomology theories: a roadmap**
“Model categories, a tour de force”
ALGANT Summer School in Brixen-Bressanone & University of Padova, Italy

PUBLICATIONS

1. **Logarithmic prismatic cohomology, motivic sheaves, and comparison theorems** (with Tommy Lundemo, Alberto Merici, and Doosung Park). (55 pages, revised 2026). [arXiv:2312.13129](#) [math.AG]. To appear in *J. Reine Angew. Math. (Crelle)*.
2. **A motivic approach to rational p -adic cohomologies** (with Alberto Vezzani). (28 pages). [arXiv:2508.16196](#) [math.AG]. To appear in *Regulators V*, Contemporary Mathematics, AMS.
3. **Logarithmic TC via the Infinite Root Stack and the Beilinson Fiber Square** (with Tommy Lundemo, Alberto Merici, and Doosung Park). (20 pages, 2024). [arXiv:2408.15627](#) [math.AG]. To appear in *Trans. Amer. Math. Soc.* [doi:10.1090/tran/9664](#).
4. **Logarithmic motivic homotopy theory** (with Doosung Park and Paul Arne Østvær). (145 pages, revised 2025). [arXiv:2303.02729](#) [math.AG]. To appear in *Mem. Amer. Math. Soc.*
5. **Motivic monodromy and p -adic cohomology theories** (with Martin Gallauer and Alberto Vezzani). (51 pages, 2024). [arXiv:2306.05099](#) [math.AG]. Published online first in *J. Eur. Math. Soc. (JEMS)* (2025).
6. **On the logarithmic slice filtration** (with Doosung Park and Paul Arne Østvær). *Geom. Topol.* 29 (2025), no. 5, 2653–2693. [arXiv:2403.03056](#) [math.AG].
7. **On the p -adic weight monodromy conjecture for complete intersections in toric varieties** (with Alberto Vezzani and Hiroki Kato). *Inventiones Math.* 241 (2025), 559–603. [arXiv:2207.00369](#) [math.AG].
8. **Logarithmic prismatic cohomology via logarithmic THH** (with Tommy Lundemo, Doosung Park and Paul Arne Østvær). *Int. Math. Res. Not. IMRN.* 2023, no. 22, 19641–19696.

9. **A Hochschild–Kostant–Rosenberg theorem and residue sequences for logarithmic Hochschild homology** (with Tommy Lundemo, Doosung Park, and Paul Arne Østvær). *Adv. Math.* 435 (2023), part A, Article 109354. [arXiv:2209.14182](#) [math.AG].
10. **Derived log Albanese sheaves** (with Alberto Merici and Shuji Saito). *Adv. Math.* 417 (2023), Article 108936. [arXiv:2107.00984](#) [math.AG].
11. **Suslin homology via cycles with modulus and applications** (with Amalendu Krishna). *Trans. Amer. Math. Soc.* 376 (2023), no. 2, 1445–1473. [arXiv:2202.06808](#) [math.AG]. [doi:10.1090/tran/8815](#).
12. **Connectivity and purity for logarithmic motives** (with Alberto Merici). *J. Inst. Math. Jussieu* 22 (2023), no. 1, 335–381. [arXiv:2012.08361](#) [math.AG]. [doi:10.1017/S1474748021000256](#).
13. **On the cohomology of reciprocity sheaves** (with Kay Rülling and Shuji Saito). *Forum Math. Sigma* 10 (2022), e72. [arXiv:2010.03301](#) [math.AG]. [doi:10.1017/fms.2022.51](#).
14. **Triangulated categories of logarithmic motives over a field** (with Paul Arne Østvær and Doosung Park). *Astérisque* 433 (2022), 280 pp. [arXiv:2004.12298](#) [math.AG].
15. **Zero cycle groups on algebraic varieties** (with Amalendu Krishna). *J. de l'Ecole Polytechnique – Mathématiques.*, 9 (2022), 281–325. [arXiv:2104.07968](#) [math.AG].
16. **Bloch’s formula for 0-cycles with modulus and higher-dimensional class field theory** (with Amalendu Krishna and Shuji Saito). *J. Algebraic Geom.* 32 (2023), no. 2, 323–384. [arXiv:2002.01856](#) [math.AG]. [doi:10.1090/jag/792](#).
17. **Rigidity for relative 0-cycles** (with Amalendu Krishna). *Ann. Sc. Norm. Super. Pisa Cl. Sci. (5)*, Volume XXII, Issue 1, pp. 241–267, 2021. [Published version](#).
18. **A Motivic homotopy theory without $\mathbb{A}^1$ -invariance**. *Math. Zeitschrift*, 295 (2020), 1475–1519 [Published version](#).
19. **Algebraic cycles with moduli and regulator maps** (with Shuji Saito). *J. Inst. Math. Jussieu* 18 (2019), no. 6, 1233–1293. [Published version](#).
20. **Zero cycles with modulus and zero cycles on singular varieties**. (with Amalendu Krishna) *Compositio Math.*, 154 (2018), no. 1, 120–187. [Published version](#).
21. **Torsion zero cycles with modulus on affine varieties**. *J. of Pure and App. Algebra*, 222 (2018), no. 1, 61–74. [Published version](#).
22. **A cycle class map for zero cycles with modulus to higher relative K-groups**. *Documenta Math.*, 23 (2018), 407–444. [Published version](#).
23. **Torsion and divisibility for reciprocity sheaves and 0-cycles with modulus**. (with Jin Cao, Wataru Kai and Rin Sugiyama) *J. of Algebra*, 469 (2017), 437–463. [Published version](#).

PREPRINTS

1. **An integral Hyodo–Kato isomorphism** (with Shuji Saito). (41 pages, 2026). [arXiv:2609.00723](#) [math.AG].
2. **On the p -adic deformation problem for the K -theory of semistable schemes** (with Tommy Lundemo, Alberto Merici, and Doosung Park). (22 pages, 2026). [arXiv:2601.14146](#) [math.AG].
3. **Descent problems for derived Azumaya algebras** (with Mauro Porta). (33 pages, Revised 2023). [arXiv:2107.03914](#) [math.AG]. Submitted.

THESIS **Motives and algebraic cycles with moduli conditions.**
 Ph.D. thesis, University of Duisburg-Essen (2016).
 DuEPublico ID: 41950. [Online version](#).

EDITORIAL ACTIVITY

Editor of the following volumes:

1. **Regulators V**
 with José Ignacio Burgos Gil, James Lewis, Deepam Patel, Gregory Pearlstein
 Contemporary Mathematics, Volume 842, American Mathematical Soc., (2026). [doi:10.1090/conm/842](#)
2. **Motivic homotopy theory and refined enumerative geometry**
 with Marc Levine, Oliver Röndigs, Toan Manh Nguyen
 Contemporary Mathematics, Volume 745, American Mathematical Soc., (2020). [doi:10.1090/conm/745](#)

RESEARCH GROUPS AND GRANTS

- PI of the special trimester program “Arithmetic Geometry”, funded by the Italian Ministry of University and Research (MUR) via the Excellence Project at the Department of Mathematics, University of Milan. 2027
Webpage: <https://sites.unimi.it/arith-geom-2027/>
- PRIN 2022 - *The arithmetic of motives and L-functions*; Italian Ministry of University and Research (MUR) 2022 – 2025
- Gruppo Nazionale per le Strutture Algebriche, Geometriche e le loro Applicazioni, INDAM 2020 – 2022
- Member of the Collaborative Research Center (SFB) 1085, “Higher Invariants”, DFG (Germany). 1 Sep 2016-
31 May 2019
Scientific project: “Cycle Classes in p-Adic Cohomology”
Coordinator: Moritz Kerz (Regensburg)
Webpage: <http://www-app.uni-regensburg.de/Fakultaeten/MAT/sfb-higher-invariants/index.php/About>
- Member of the Special Priority Program (SPP) 1786, “Homotopy Theory and Algebraic Geometry”, DFG (Germany). 01 Oct 2015-
30 Aug 2016
Scientific project: “Motives with modulus”.
Coordinators: Hélène Esnault (Berlin), Marc N. Levine (Essen), Moritz Kerz (Regensburg).
Webpage: https://www.uni-due.de/~bm0032/SPP1786/Web/projects_EsnaultKerzLevine

SHORT RESEARCH VISITS

Sept 2024	University of Tokyo, Japan Host: Shuji Saito
Sept 2023	University of Tokyo, Japan Host: Shuji Saito
Dec 2019	ETH Zürich, Switzerland Host: Joseph Ayoub
Mar 2019	Oslo University, Norway Host: P.-A. Østvær
Apr 2018	Tohoku University, Japan Host: T. Yamazaki
Mar 2018	Oslo University, Norway Host: P.-A. Østvær
Dec 2017	Tohoku University, Japan Host: T. Yamazaki
Feb 2017	Tohoku University, Japan Host: T. Yamazaki

SUMMER SCHOOLS AND THEMATIC WORKSHOPS (SELECTION)

Mar 2026	Simons collaboration on Perfection in Algebra, Geometry and Topology Annual Meeting (Invited participant) Simons Foundation, NY, USA
Mar 2025	Simons collaboration on Perfection in Algebra, Geometry and Topology Annual Meeting (Invited participant) Simons Foundation, NY, USA
Nov 2023	Workshop on p-adic arithmetic geometry Organizers: Bhargav Bhatt (IAS/Princeton/Michigan), Kestutis Cesnavicius (CNRS Université Paris-Saclay), Jacob Lurie (IAS) IAS, Princeton, USA
July 2023	Recent advances in algebraic K-theory Organizers: Benjamin Antieau (Northwestern University), Lars Hesselholt (University of Copenhagen / Nagoya University), and Matthew Morrow (CNRS and Université Paris-Saclay) IHES, France
Nov 2020	Masterclass in Condensed Mathematics. Lecture series by Peter Scholze and Dustin Clausen. Organizers: Lars Hesselholt, Ryomei Iwasa, Markus Land, Piotr Stragowski. University of Copenhagen, Denmark (Virtual)
Jun 2019	Workshop Algebraic K-theory. Organizers: Thomas Geisser, Lars Hesselholt, Annette Huber-Klawitter, Moritz Kerz. Mathematisches Forschungsinstitut Oberwolfach (MFO), Oberwolfach, Germany
Sep 2018	Motives in St. Petersburg. Summer school “Motives and related structures”. Organizers: Alexey Ananyevskiy, Ivan Panin, Grigory Garkusha. Euler International Mathematical Institute, St. Petersburg, Russia

- Jul 2016* **Algebraic K -theory and motivic cohomology**
Organizers: Thomas Geisser, Annette Huber-Klawitter, Uwe Jannsen, Marc Levine
Mathematisches Forschungsinstitut Oberwolfach (MFO), Oberwolfach, Germany
- Aug 2015* Summer school **Algebraic K -theory and trace methods**. Lecture series by Lars Hesselholt e Chuck Weibel.
University of Regensburg, Regensburg, Germany.
- Jul 2015* **AMS Summer Institute (Algebraic Geometry)**
Organizers: Hélène Esnault, Mark Kisin, Peter Scholze, Shou-Wu Zhang
University of Utah, Salt Lake City, USA.
- Mar 2015* School “ **p -adic arithmetic**”. Lecture series by Gabriel Dospinescu, Laurent Fargues, e Eike Lau.
University of Duisburg-Essen, Essen, Germany.
- Nov 2014* **International Workshop on motives in Tokyo**
University of Tokyo, Tokyo, Japan
- Oct 2014* Summer school **Continuous K -theory of p -adic rings (after Beilinson)**
Organizers: Hélène Esnault, Moritz Kerz
Universität Regensburg, Germany at Marburger Haus, HirscheggKleinwalsertal, Austria
- Nov 2013* **International Workshop on motives in Tokyo**
University of Tokyo, Tokyo, Japan
- Sep 2013* Summer school **Motivic Galois group**. Lectures by Joseph Ayoub, Yves André and Marc Levine
Alfréd Rényi Institute of Mathematics, Budapest, Hungary

TEACHING

- | | |
|---|---|
| ▪ Algebra 4
Laurea triennale in Matematica (9 CFU)
Lectures and exercises, 54 hours (42+12 hours). | <i>Spring 2026</i>
<i>Università degli Studi di Milano</i> |
| ▪ Commutative Algebra
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures, 42 hours. | <i>Fall 2025</i>
<i>Università degli Studi di Milano</i> |
| ▪ Algebra 2
Laurea triennale in Matematica (6 CFU)
Lectures, 30 hours. | <i>Fall 2025</i>
<i>Università degli Studi di Milano</i> |
| ▪ Derived Geometry
Ph.D. course
Lectures, 15 hours. | <i>Spring 2025</i>
<i>Università degli Studi di Milano</i> |
| ▪ Algebra 4
Laurea triennale in Matematica
(6 CFU) Lectures, 36 hours. | <i>Spring 2025</i>
<i>Università degli Studi di Milano</i> |
| ▪ Commutative Algebra
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures, 42 hours. | <i>Fall 2024</i>
<i>Università degli Studi di Milano</i> |
| ▪ Algebra 2
Laurea triennale in Matematica (6 CFU)
Lectures, 27 hours. | <i>Fall 2024</i>
<i>Università degli Studi di Milano</i> |
| ▪ Homological Algebra
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures, 14 hours. | <i>Spring 2024</i>
<i>Università degli Studi di Milano</i> |
| ▪ Algebra 4
Laurea triennale in Matematica
(6 CFU) Exercises, 24 hours. | <i>Spring 2024</i>
<i>Università degli Studi di Milano</i> |
| ▪ Homological Algebra
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures, 21 hours. | <i>Spring 2023</i>
<i>Università degli Studi di Milano</i> |
| ▪ Algebra 2 | <i>Fall 2022</i> |

- Laurea triennale in Matematica (6 CFU)
Exercises, 24 hours. *Università degli Studi di Milano*
- **Commutative Algebra** *Fall 2022*
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures and exercises, 19 hours. *Università degli Studi di Milano*
 - **Homological Algebra** *Spring 2022*
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures, 21 hours. *Università degli Studi di Milano*
 - **Algebra 2** *Fall 2021*
Laurea triennale in Matematica (6 CFU)
Exercises, 24 hours. *Università degli Studi di Milano*
 - **Commutative Algebra** *Fall 2021*
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures and exercises, 19 hours. *Università degli Studi di Milano*
 - **Homological Algebra** *Spring 2021*
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures, 21 hours. *Università degli Studi di Milano*
 - **Commutative Algebra** *Fall 2020*
Master ALGANT / Laurea Magistrale in Matematica (6 CFU)
Lectures and exercises, 19 hours. *Università degli Studi di Milano*
 - **Matematica del Discreto** *Fall 2019*
Laurea triennale in Sistemi e Sicurezza delle Reti Informatiche
(6 CFU) Lectures and exercises, 16 hours. *Università degli Studi di Milano*
 - **Commutative Algebra** *Fall 2019*
Master ALGANT / Laurea Magistrale in Matematica (9 CFU)
Lectures and exercises, 27 hours. *Università degli Studi di Milano*
 - **The Grothendieck-Riemann-Roch theorem** *Spring 2018*
Master and PhD course (3 ECTS)
Lecturer (in collaboration with Adeel Khan (Uni Regensburg))
Webpage: <https://www.preschema.com/teaching/grr-ss18> *Universität Regensburg (Germany)*
 - **Zero cycles on singular varieties** *04/2017 - 07/2017*
Master and PhD course (12 hours) *Universität Regensburg (Germany)*
 - **Algebraic Geometry III** *10/2015-02/2016*
Bachelor in Mathematics (26 hours) *Universität Duisburg-Essen (Germany)*
Teaching assistant (lecturer A. Rodriguez)
Program: abelian varieties, algebraic and complex theory
Webpage: <https://www.esaga.uni-due.de/marc.levine/Courses/2015/AlgGeom3/>
 - **Commutative Algebra** *06/10/2015-16/10/2015*
Intensive course for master's students (2 weeks, 28 hours) *Universität Duisburg-Essen (Germany)*
Lecturer
Webpage: <https://www.esaga.uni-due.de/marc.levine/Courses/2015/ComAlgShort/>
 - **Algebra II** *04/2015 - 07/2015*
Bachelor in Mathematics (26 hours) *Universität Duisburg-Essen (Germany)*
Teaching assistant (lecturer Prof. Marc Levine)
Program: ring theory, modules, elements of affine algebraic geometry
Webpage: <https://www.esaga.uni-due.de/marc.levine/Courses/2015/Algebra2/>
 - **Algebra I** *04/2011 - 07/2011*
Laurea triennale in Matematica *Università degli Studi di Milano*
Tutoring (lecturer Prof. Luca Barbieri-Viale)

STUDENT SUPERVISION

Ph.D. thesis advisor
Student: Eliot Hecky

(Started) 2024
University of Milano

Ph.D. thesis advisor (Started) 2024
 Student: Xiaomin Chu University of Milano

Ph.D. thesis advisor (2023 - 2026)
 Student: Ruichuan Zhang University of Milano
 Title: Moduli stacks of derived log structures

Master thesis advisor 2026
 Student: Gabriele Olivieri Università degli Studi di Milano

Master thesis advisor 2026
 Student: Alessandra Argentino Università degli Studi di Milano

Master thesis advisor 2021
 Student: Ottavio Bartenor Università degli Studi di Torino
 Co-supervision with Prof. Elena Martinengo

Master thesis advisor 2022
 Student: Luca Passolunghi Università degli Studi di Milano
 Co-supervision with Prof. Alberto Vezzani

POSTDOC SUPERVISION

Postdoc supervision 2024-2025
 Dr. Raphaël Ruimy Università degli Studi di Milano
 (joint with F. Andreatta)

ORGANIZATION OF WORKSHOPS & SEMINARS

- September 2025* “Yatsugatake workshop 2025: Wild ramification and Etale homotopy, Haramura, Nagano, Japan. Financed by JSPS, ERC. With Piotr Achinger (Warsaw), Ko Aoki (MPIM), Wataru Kai (Tohoku), Alberto Merici (Heidelberg), Hiroyasu Miyazaki (NTT), Shuji Saito (Tokyo)
- January 2025* “Algebraic geometry: a motivic view. Conference in honour of Marc Levine”, ETH Zurich, Switzerland. Financed by ETH, K-Theory Foundation, [Conference website](#). With Aravind Asok (USC), Joseph Ayoub (UZurich), Elden Elmanto (Toronto), Maria Yakerson (Oxford)
- September 2024* “Yatsugatake workshop 2024: Chromatic Homotopy Theory”, Haramura, Nagano, Japan. Financed by JSPS, K-Theory Foundation, Foundation Nagoya Math. Journal. [Conference website](#). With Ko Aoki (MPIM), Ryomei Iwasa (Orsay), Alberto Merici (Heidelberg), Hiroyasu Miyazaki (NTT), Shuji Saito (Tokyo)
- June 2024* “Regulators V - a conference in honor of Spencer Bloch and Rob de Jeu”, University of Pisa, Italy. with J. Lewis, D. Patel, G. Pearlstein, T. Szamuely. Financed by Compositio Foundation, K-theory Foundation, NSF, Clay Math. Institute, INdAM. The proceedings of the conference will be published by the AMS.
- Sep 2023* “Yatsugatake workshop 2023: Condensed Mathematics and analytic K-theory”, Haramura, Nagano, Japan. Financed by JSPS (Grant Prof. S. Saito and Prof. T. Terasoma). [Conference website](#).
- Sep 2023* “Giovani ricercatori in algebra e geometria”, Università degli Studi di Roma Tor Vergata, Roma. Financed by INDAM.
- Feb 2022* “Motives from 1 to 62”. Università degli Studi di Milano. [Conference site](#) Financed by Compositio Foundation, K-Theory Foundation, LYSM. With F. Andreatta and A. Vezzani
- Sep 2021* “The six-functor formalism and motivic homotopy theory”. Università degli Studi di Milano. Financed by the Research Council of Norway. With A. Vezzani and Paul Arne Østvær.
- 2-6 Sep 2019* “Yatsugatake workshop 2019: Derived algebraic geometry”, Haramura, Nagano, Japan. Financed by JSPS (Grant Prof. S. Saito and Prof. T. Terasoma). [Conference website](#).
- 14-18 May 2018* International workshop “Motives and refined enumerative geometry”, Universität Duisburg-Essen (Germany). Financed by the DFG, project SPP 1786. [Conference website](#). The proceedings of the conference have been published by the AMS.
- fall 2015* Seminar of the algebraic and arithmetic geometry’s research group, Universität Duisburg-Essen (Germany). Topic: “Class Field theory for varieties over finite fields”. [Link to official program](#).

REFEREING AND EVALUATION ACTIVITIES

- **Journal refereeing.** Full referee reports or short evaluations for the following journals: *Annals of Mathematics*, *Acta Mathematica*, *Journal of the American Mathematical Society*, *Inventiones Mathematicae*, *Journal of the European Mathematical Society*, *Journal für die reine und angewandte Mathematik (Crelle)*, *Advances in Mathematics*, *Journal of the Institute of Mathematics of Jussieu*, *Selecta Mathematica*, *International Mathematics Research Notices*, *Nagoya Mathematical Journal*, *Mathematical Research Letters*, *Canadian Journal of Mathematics*, *Canadian Mathematical Bulletin*, *Documenta Mathematica*, *Épjournal de Géométrie Algébrique*, *Homology, Homotopy and Applications*, *Annals of K-Theory*, and *Annali della Scuola Normale Superiore di Pisa*.
- **Mathematical Reviews.** Reviewer for *Mathematical Reviews* (American Mathematical Society), until 2024.
- **Thesis and grant evaluation.**
 - Referee for the Ph.D. thesis of Sofian Tur-Dorvault (Université de Montpellier, France, 2026).
 - Referee for the Ph.D. thesis of Sandeep R. (Tata Institute of Fundamental Research, India, 2026).
 - Referee for the Ph.D. thesis of Xinyu Shao (Sorbonne Université, France, 2025).
 - Referee for the Ph.D. thesis of Mainak Gosh (Tata Institute of Fundamental Research, India, 2021).
 - Referee for the Ph.D. thesis of Gretar Amazeen (University of Wuppertal, Germany, 2020).
 - External evaluator of grant proposals for the Deutsche Forschungsgemeinschaft (DFG).
 - External evaluator of grant proposals for the Agence nationale de la recherche (ANR).

RECOGNITION

- Elected member of the K -theory foundation September 2023
- Admitted to the final step of the ERC Starting Grant 2023 MOTIVIC LOGARITHMIC GEOMETRY (MOLOGE). Final panel score: A (fully meets the ERC’s excellence criterion and is recommended for funding if sufficient funds are available)

SERVICE

- Math. Department representative in the Open Science committee, University of Milano 2023 –
- Gender Equality Committee, Dipartimento di Matematica “F.Enriques”, Unimi. 2025 –
- (Previously) Math. Department representative in the Academic Committee for fixed term researchers (Università degli Studi di Milano). 2023 – 2024

OUTREACH AND PUBLIC ENGAGEMENT

Outreach publications:

- **How to apply for a postdoc in Europe?** with Chiara Damiolini. Notices of the American Mathematical Society 67 (10), 1563-1565 (2020)
- **Is there a Human Right to Open Science? (Il diritto umano alla scienza aperta)** with Roberto Caso. - [s.l.] : Università di Trento, (2020) (TRENTO LAW AND TECHNOLOGY RESEARCH GROUP RESEARCH PAPER SERIES) [10.5281/zenodo.4053531]
- **Il diritto umano alla scienza aperta (the human right to open science)**, con Roberto Caso - appeared in the volume “Così san tutt3”, Fandango Libri, 2021. A cura di Marco Perduca e Giulia Perrone. Prefazione di Chiara Valerio.

Other outreach/public engagement activities:

- Advisor role for COPER (Consulta degli Enti pubblici di Ricerca, national Council on Research Agencies) for Open Science, coordinated by INFN (Istituto Nazionale di Fisica Nucleare), since 2022.
- Member of the working group in the Italian Chapter of CoARA (Coalition for Advancing Research Assessment).

QUALIFICATIONS

- **Maître de conférences** Obtained in 2017
Conseil National des Universités (CNU), France, section 25 (pure mathematics)
- **Abilitazione Scientifica Nazionale, seconda fascia** 03/07/2020 – 03/07/2029
Bando DD 2175/2018, S.C. 01/A2 Geometria e Algebra
- **Abilitazione Scientifica Nazionale, prima fascia** 24/10/2023 – 24/10/2034
Bando DD 553/2021, S.C. 01/A2 Geometria e Algebra